Severe Traumatic Brain Injury

All patients with:

Risk of intracranial hypertension (ICP$\geq$ 22 mmHg) OR
GCS $\leq$ 9 and an abnormal CT scan OR
GCS $\leq$ 9 with a normal CT and 2 of the following

a. Systolic blood pressure <90
b. Age > 40
c. Decorticate or decerebrate posturing

Admit to

1. Isolated severe brain injuries will be admitted directly to the Neuro ICU under neurosurgery. Trauma service will complete a tertiary survey at 24 hours.
2. Polytrauma patients will be admitted to the trauma service in the trauma ICU where the patients will still receive high level neurotrauma care.

Initial Consultations- to be placed by admitting service, as needed

1. Neurosurgery
2. Neurocritical Care
3. Trauma surgery

Must have the following on admission to ICU

1. ICP Monitor placed within 6 hours, parenchymal or EVD
2. Cerebral oximetry monitor placed within 6 hours, parenchymal
3. Invasive arterial monitoring
4. Bladder temperature or core temperature monitor
5. End Tidal CO_2 monitor
6. 24 hour EEG within 24 hours of admission (order: continuous long term video EEG)
7. Central venous line, as necessary (3% NaCl > 50mL/hr, 23% NaCl and/or Vasopressors)
8. Antiepileptic prophylaxis

   Keppra IV 1000mg loading dose followed by 750mg Q12h for 7 days

   *Variation in antiepileptic therapy at the discretions of neurocritical care and/or neurosurgery
9. VAP prophylaxis

   Ceftriaxone 2g for 1 dose on admission or within 12 hrs of intubation

Initial management goals

1. Maintain CPP $\geq$ 60
2. Avoid fevers. Maintain temperature $\leq$37.5°C
3. ICP$\leq$ 22 mmHg
4. $PbtO_2 \geq 20$ mmHg, if cerebral oxygen is being monitored
5. Target CO_2 35-40 mmHg
6. Sodium goals of 145-150
7. Avoid hypoxia. Maintain $SpO_2 > 90$ and $PaO_2 > 80$ mmHg
8. Hemoglobin > 7 g/dL
9. Blood glucose per ICU protocol 100-180mg/dL
10. Ensure adequate sedation to achieve ICP goals

Algorithms:

Hypotension- CPP< 60 mmHg or MAP< 65mmHg

1. Fluid resuscitation
    a. Evaluate volume status with the goal of euvolemia
    b. Serial bedside POCUS as indicated
    c. Fluid replacements
        i. 0.5 to 1L IV 0.9% saline or plasmalyte bolus
        ii. 0.9% saline or plasmalyte maintenance infusion
    d. If Hgb <7g/dL, transfuse
2. Vasopressor therapy
    a. Norepinephrine 0.01-1 mcg/kg/min and/or
    b. Phenylephrine 0.5-5 mcg/kg/min and/or
    c. Vasopressin 0.03 U/min (non-titratable)
3. Inotropic therapy- obtain critical care attending approval prior to initiation
    a. Epinephrine 0.01-1 mcg/kg/min and/or
    b. Milrinone 0.125-0.5 mcg/kg/min

High ICP and Normal or unmeasured cerebral oxygen (ICP ≥ 22 mmHg and PbtO2 ≥20 mmHg)

Tier 1: Start Tier 1 therapies within 15 min of the start of the episode of elevated ICP as detected by continuous ICP recordings.

1. Assure Head of bed is 30-45° (when cleared off log roll, otherwise reverse trendelenburg) and the head is in neutral position
2. Initiate fever suppression measures if temperature >37.5° C
3. Adjust sedation and analgesic infusions to ICP goals (not RASS) and bolus if needed
    Infusions:
        Propofol 5-50 mcg/kg/min infusion and/or
        Fentanyl 25-300 mcg/h infusion or
        Hydromorphone 0.1-4 mg/h infusion and/or
        Midazolam 2-12 mcg/h infusion and/or
        Ketamine 0.3-5 mg/kg/h infusion with an initiation bolus of 1.5-mg/kg
    Bolus:
        Propofol- 50-100 mg IV or
        Ketamine 100 mg IV or
        Fentanyl 50-100 mcg IV or
        Hydromorphone 0.4-1mg IV or
        Midazolam 5 mg IV
4. Hypertonic therapy
    bolus hypertonic fluid if below sodium goals. *Use hypertonic orderset*
        30 mL of 23.4% NaCl over 30 min or
        300 mL of 3% NaCl over 20 min
    Can consider sodium acetate if hyperchloremic
5. Hyperosmotic therapy

Bolus mannitol 0.5g/kg
Ensure serum Osm <320 and Osm gap <20
Ensure euvolemia- monitor uop and replace with isotonic fluids
Caution in patients undergoing active resuscitation for shock

6. CSF drainage- contact primary, neurosurgery and neurocritical care
   Drainage to be done by trained beside nurse through EVD if present
   Open EVD to 0 and drain 5 mL of CSF over 1 min. Stop at one min or 5mL whichever occurs first, unless directed otherwise by neurosurgery.
7. Consider EEG to rule out underlying seizure
8. Obtain repeat head imaging to evaluate for worsening mass lesion and midline shift

Tier 2: Move to Tier 2 interventions if ICP >22 mmHg for > 60 min despite Tier 1 therapies
Upon initiating Tier 2 therapies, primary, neurosurgery, and neurocritical care should be notified

1. Temporarily reduce pCO2 goal to 32-35 mmHg
2. Hyperosmotic therapy
   Bolus higher dose mannitol 1-2g/kg
   Ensure serum Osm <320 and Osm gap <20
   Ensure euvolemia- monitor uop and replace with isotonic fluids
   Caution in patients undergoing active resuscitation for shock
3. Target temperature management
   Consider targeting 35-37°C. Temperature management at the discretion of the treating teams
4. Initiate neuromuscular paralysis
   Titrate to 1 out of 4 twitches
   Vecuronium 0.1mg/kg bolus followed by 0.05-1.2mcg/kg/min infusion or
   Cisatracurium 200mcg/kg bolus followed by 3mcg/kg/min infusion
5. Trial of CPP goal ≥70 mmHg
   Vasopressors as above
6. Consider repeat head imaging to evaluate for worsening mass lesion and midline shift
7. Consider EEG to rule out underlying seizure

Tier 3: Move to Tier 3 if elevated ICP remain despite Tier 1 and Tier 2 therapies.
Prior to initiating Tier 3 therapies, primary, neurosurgery, and neurocritical care must be notified

1. Anesthetic Burst Suppression
   Optimize sedation/ analgesia for burst suppression on continuous EEG
2. Consider Pentobarbital Coma
   Use *adult collaborative pentobarbital order set*
   Load: 5mg/kg IV over 30 min Followed by 15mg/kg IV over three hours.
   Maintenance: 1 mg/kg/hr infusion. Titrate to ICP and EEG burst suppression goal
   All patients should have cardiac output monitoring
   Patients will remain suppressed for 72 hours. After which the pentobarbital will be discontinued.
   NO brain death testing, including clinical and all ancillary exams, can be done until pentobarbital serum levels are < 5µg/mL. (Around 5 days)
2. Consider decompressive craniotomy
3. Consider lowering temperature target

Temperature goals maybe lowered to 32-35°C for a maximum duration of 72 hours followed by rewarming at a rate of 0.25-0.5°C/hr

Changes in temperature goals must be approved by admitting surgical team given possibilities of cold induced coagulopathy, as well as the critical care, neurosurgical, and neurocritical care teams.

4. Consider repeat head imaging to evaluate for worsening mass lesion and midline shift
5. Consider EEG to rule out underlying seizure

Low ICP and Low Cerebral oxygen ICP<22 mmHg and PbtO2<20 mmHg

Tier 1: Start Tier 1 therapies within 15min of the start of the episode of elevated ICP as detected by continuous ICP recordings.

1. Assure Head of bed is 30-45° (when cleared off log roll, otherwise reverse trendelenburg) and the head is in neutral position
2. Initiate fever suppression measures if temperature >37.5° C
3. Increase CPP goal to 70 mmHg
   Trial of fluid bolus to achieve new goal- 1L 0.9% saline or plasmalyte
   Increase MAP goals to 70 mmHg and use vasopressors if needed
4. Consider EEG to rule out underlying seizure
5. Adjust ventilator parameters to increase PaO2.
   a. Increase FiO2 to 60%
      If PbtO2 does not increase ≥ 5 mmHg within 2 hours return to prior FiO2
      If PbtO2 increases ≥ 5 mmHg, maintain FiO2 for 6 hours and then return to prior FiO2. Reassess
   b. Consider increasing PEEP to increase PaO2

Tier 2: Move to Tier 2 interventions if PbtO2 < 20 mmHg for > 60 min despite Tier 1 therapies
Upon initiating Tier 2 therapies, primary, neurosurgery and neurocritical care should be notified

1. Increase CPP goal to > 70 mmHg and use vasopressors to achieve the goal
2. Adjust minute ventilation to increase PaCO2 to 40-45 mmHg
   This will increase cerebral blood flow if the auto regulatory reflexes are intact.
3. Decrease ICP to goal of ≤10 mmHg
   a. CSF drainage- contact primary, neurosurgery and neurocritical care
      Drainage to be done by trained beside nurse through EVD if present
      Open EVD to 0 and drain 5 mL of CSF over 1 min. Stop at one min or 5mL whichever occurs first, unless directed otherwise by neurosurgery.
   b. Increase sedation
4. Consider EEG to rule out underlying seizure
5. Adjust ventilator parameters to increase PaO2.
   a. Increase FiO2 to 100%
      If PbtO2 does not increase ≥ 5 mmHg within 2 hours return to prior FiO2
      If PbtO2 increases ≥ 5mmHg, maintain FiO2 for 6 hours and then return to prior FiO2. Reassess
   b. Consider increasing PEEP to increase PaO2

High ICP and Low Cerebral oxygen (ICP ≥ 22 mmHg and PbtO2 < 20 mmHg)

Tier 1: Start Tier 1 therapies within 15 min of the start of the episode of elevated ICP as detected by continuous ICP recordings.

1. Assure Head of bed is 30-45° (when cleared off log roll, otherwise reverse trendelenburg) and the head is in neutral position
2. Initiate fever suppression measures if temperature >37.5° C
3. Adjust sedation and analgesic infusions to ICP goals (not RASS) and bolus if needed
    Infusions:
        Propofol 5-50 mcg/kg/min infusion and/or
        Fentanyl 25-300 mcg/hr infusion or
        Hydromorphone 0.1-4 mg/hr infusion and/or
        Midazolam 2-12 mcg/hr infusion and/or
        Ketamine 0.3-5 mg/kg/hr infusion with an initiation bolus of 1.5-mg/kg
    Bolus:
        Propofol- 50-100 mg IV or
        Ketamine 100 mg IV or
        Fentanyl 50-100 mcg IV or
        Hydromorphone 0.4-1mg IV or
        Midazolam 5 mg IV or
4. CSF drainage- contact primary, neurosurgery, and neurocritical care
    Drainage to be done by trained beside nurse through EVD if present
    Open EVD to 0 and drain 5 mL of CSF over 1 min. Stop at one min or 5mL whichever occurs first, unless directed otherwise by neurosurgery.
5. Increase CPP goal to 70 mmHg
    Trial of fluid bolus to achieve new goal- 1L 0.9% saline or plasmalyte
6. Hypertonic therapy
    bolus hypertonic fluid if below sodium goals. *Use hypertonic orderset*
        30 mL of 23.4% NaCl over 30 min or
        300 mL of 3% NaCl over 20 min
    Can consider sodium acetate if hyperchloremic
7. Hyperosmotic therapy
    Bolus mannitol 0.5 g/kg
    Ensure serum Osm <320 and Osm gap <20
    Ensure euvolemia- monitor uop and replace with isotonic fluids
    Caution in patients undergoing active resuscitation for shock
8. Consider EEG to rule out underlying seizure
9. Adjust ventilator parameters to increase PaO2.
    a. Increase FiO2 to 60%
        If PbtO2 does not increase ≥ 5 mmHg within 2 hours return to prior FiO2
        If PbtO2 increases ≥ 5mmHg, maintain FiO2 for 6 hours and then return to prior FiO2. Reassess
    b. Consider increasing PEEP to increase PaO2

Tier 2: Move to Tier 2 interventions if ICP >22mmHg and PbtO2 < 20 mmHg for longer than 60min despite Tier 1 therapies

Upon initiating Tier 2 therapies, primary, neurosurgery, and neurocritical care should be notified

1. Consider temperature goal of 35-37°C and avoid < 34° C.
    temperature goals at the discretion of the treatment teams
2. Hyperosmotic therapy
    Bolus higher dose mannitol 1-2g/kg
    Ensure serum Osm <320 and Osm gap <20
    Ensure euvolemia- monitor uop and replace with isotonic fluids
    Caution in patients undergoing active resuscitation for shock
3. Increase CPP goal to ≥ 70 mmHg and use vasopressors to achieve the goal
4. If Hgb <7g/dL, transfuse
5. Initiate neuromuscular paralysis
    Titrate to 1 out of 4 twitches
    Vecuronium 0.1 mg/kg bolus followed by 0.05-1.2 mcg/kg/min infusion or
    Cisatracurium 200 mcg/kg bolus followed by 3 mcg/kg/min infusion
6. Adjust ventilator parameters to increase PaO2.
    a. Increase FiO2 to 100%
        If PbtO2 does not increase ≥ 5 mmHg within 2 hours return to prior FiO2
        If PbtO2 increases ≥ 5 mmHg, maintain FiO2 for 6 hours and then return to prior FiO2. Reassess
    b. Consider increasing PEEP to increase PaO2
7. Obtain repeat head imaging to evaluate for worsening mass lesion and midline shift
8. Treat surgically remediable lesions with craniotomy according to guidelines
9 . Consider EEG to rule out underlying seizure

Tier 3: Move to Tier 3 if elevated ICP remain despite Tier 1 and Tier 2 therapies.
Prior to initiating Ttier 3 therapies, primary, neurosurgery and neurocritical care must be notified

1. Anesthetic Burst Suppression
    Optimize sedation/ analgesia for burst suppression on continuous EEG
2. Consider Pentobarbital Coma
    Use *adult collaborative pentobarbital order set*
    Load: 5 mg/kg IV over 30 min Followed by 15 mg/kg IV over three hours.
    Maintenance: 1 mg/kg/hr infusion. Titrate to ICP and EEG burst
    All patients should have cardiac output monitoring
    Patients will remain suppressed for 72 hours. After which the pentobarbital will be discontinued.
    NO brain death testing, including clinical and all ancillary exams, can be done until pentobarbital serum levels are < 5μg/mL (Around 5 days)
2. Consider decompressive craniotomy
3. Consider lowering temperature target
    Temperature goals maybe lowered to 32-35°C for a maximum duration of 72 hours followed by rewarming at a rate of 0.25-0.5°C/hr
    Changes in temperature goals must be approved by admitting surgical team given possibilities of cold induced coagulopathy, as well as the critical care, neurosurgical, and neurocritical care teams.
4. Consider repeat head imaging to evaluate for worsening mass lesion and midline shift
5. Consider EEG to rule out underlying seizure

Abbreviations and formulas:

CPP- cerebral perfusion pressure
ICP- intracranial pressure
MAP- mean arterial pressure
PbtO2- brain tissue oxygen tension
PaO2- arterial partial pressure of oxygen
PaCO2- arterial partial pressure of carbon dioxide
CPP: Cerebral perfusion pressure = mean arterial pressure - intracranial pressure
Osm gap= measured osmolality- calculated osmolality

can be skewed by alcohol

Calculated Osmolality = (2 * Serum Na) + (Serum Glucose/18) + (Serum BUN/2.8)